

Defending Against Transfer-Based Adversarial Attacks Using SVD-Driven Feature Evolution

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Abstract

Due to their high stealthiness and difficulty in detection, the transfer-based adversarial attacks pose a significant challenge to the security and robustness of computer vision models. In this paper, we propose a plug-and-play **SVD-driven feature evolution module (SDFEM)** to assist image classification models in defending against transfer-based adversarial attacks. The SDFEM consists of “feature concatenation,” “feature reconstruction,” and “feature weight optimization.” After the adversarial examples are decomposed into singular value features using Singular Value Decomposition (SVD), the above three components sequentially achieve the concatenation of features along the channel dimension, the reconstruction of multi-level feature representations, and the optimization of feature weights based on channel context, thereby suppressing the features that significantly contribute to adversarial attacks. Extensive experiments demonstrate that the SDFEM effectively defends against various types of transfer-based attacks, achieving state-of-the-art black-box robustness. The code is available at <https://github.com/X-L-Liu/SDFEM>.

1. Introduction

In recent years, deep learning has been widely applied to various visual tasks [1–3] such as image classification [4, 5], object detection [6, 7], and semantic segmentation [8, 9]. However, recent studies indicate that there exist adversarial examples that do not affect human cognition but can mislead deep learning models [10–13]. Adversarial examples are generated by introducing subtle perturbations to clean examples, resulting in highly confident yet entirely incorrect predictions from well-trained classifiers. For instance, before the introduction of adversarial perturbations, a deep learning classification model correctly identifies the clean example in Fig. 1 as a “tank” with 82.4% confidence; after

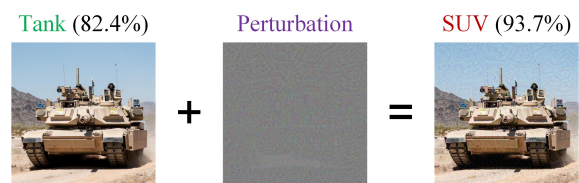


Figure 1. Clean example and adversarial example. When adversarial perturbations are added to the “tank,” it is misclassified by the classification model as an “SUV” with a higher confidence level.

the addition of adversarial perturbations, the same model incorrectly classifies it as an “SUV” with even higher confidence (93.7%). Defending against adversarial attacks has emerged as one of the significant challenges in ensuring the security of vision models [14, 15].

Transfer-based adversarial attacks [16–19] are a typical black-box attack approach where the attacker lacks access to the target model’s internal details and can only observe its input-output behavior a limited number of times. These attacks generate adversarial examples on a substitute model to attack the target model, thereby eliminating the need for detailed knowledge of the target model or multiple queries for its output results. This characteristic enhances their stealthiness, making them among common forms of adversarial attack methods. In response, adversarial training [12, 20] aims to enhance the robustness of models and optimize their decision boundaries to defend against the most severe white-box attacks (gradient-based attacks). In contrast, input transformation defenses [15, 21, 22] employ techniques such as random transformations and gradient masking to obscure the decision boundaries of the classification model. Compared to adversarial training, input transformation methods offer a more effective defense against stealthy transfer-based attacks without significantly impacting the recognition rate of clean examples [23, 24].

Image decomposition operations, such as Fourier Transform [25], Discrete Cosine Transform [26], and Singular

Value Decomposition (SVD) [27], are frequently employed to defend against adversarial attacks [28–30]. In this process, the defender disrupts the perturbation structure within adversarial examples by removing certain frequency components. However, traditional methods exhibit poor specificity, failing to accurately identify and remove the components that carry adversarial characteristics, resulting in low recognition rates for both clean and adversarial examples.

To address the above issues, we propose a plug-and-play *SVD-driven feature evolution module* (SDFEM) to suppress singular value features that contribute significantly to adversarial attacks. This module comprises three components: “feature concatenation,” “feature reconstruction,” and “feature weight optimization.” Adversarial examples are first processed by the feature concatenation component, transforming them into a set of singular value features concatenated along the channel dimension. The feature reconstruction component then merges and deepens the singular value features, endowing them with diverse feature representations. Next, the feature weight optimization component calculates feature weights based on a channel attention mechanism to suppress those features that carry more adversarial characteristics. Finally, the reconstructed and optimized singular value features are subsequently fed into the target model for classification. Compared to previous defense strategies, SDFEM demonstrates superior performance in countering transfer-based adversarial attacks.

We summarize the main contributions as follows:

- We propose a plug-and-play *SVD-driven feature evolution module* (SDFEM) that effectively suppresses the singular value features in adversarial examples which significantly contribute to adversarial attacks, thereby defending against transfer-based attacks and enhancing the black-box robustness of classification models.
- We validate the contributions of different singular value features to adversarial attacks and find significant differences in the distribution of these contributions across different categories, indicating that previous fixed feature removal methods are indeed inadequate.
- The evaluation indicates that SDFEM achieves a balance between natural and robust accuracies, demonstrating superior defensive performance against transfer-based attacks and significantly outperforming previous state-of-the-art defense approaches in black-box scenarios.

2. Related Work

2.1. Adversarial Example

The generation of adversarial examples is a constrained optimization framework. Consider a pre-trained classifier $\mathcal{C}(\cdot)$ mapping inputs $\mathbf{x} \in \mathbb{R}^m$ to class labels $\ell \in \mathbb{Z}^+$. Let an attack algorithm $\mathcal{A}(\cdot)$ craft norm-bounded perturbations $\boldsymbol{\rho} \in \mathbb{R}^m$ under parameter $\boldsymbol{\theta}$, such that $\mathcal{A}(\boldsymbol{\theta}, \mathbf{x}) \rightarrow \boldsymbol{\rho}$.

To preserve human-interpretable semantics, the perturbation magnitude is constrained by $\|\boldsymbol{\rho}\|_p < \epsilon$, where ℓ_p norm bounds $p \in \{0, 1, 2, \infty\}$ and ϵ denotes the adversarial budget. The adversarial example $\bar{\mathbf{x}}$ is then synthesized via $\bar{\mathbf{x}} = \mathbf{x} + \boldsymbol{\rho}$. This process is formally expressed as:

$$\arg \max_{\boldsymbol{\rho}} \mathcal{L}(\boldsymbol{\theta}, \bar{\mathbf{x}}, \ell) \quad \text{s.t.} \quad \|\boldsymbol{\rho}\|_p < \epsilon, \quad (1)$$

where $\mathcal{L}$ quantifies the prediction error of the model, commonly implemented through cross-entropy. The objective is to minimize the performance of the classifier as much as possible within the established perturbation range.

2.2. Transfer-Based Adversarial Attacks

Transfer-based adversarial attacks improve the transferability of adversarial examples by transforming images, optimizing gradient variances, integrating attack techniques, and among other strategies. Fast gradient sign method (FGSM) generates the adversarial perturbations by following the gradient direction of the loss function with respect to the input image [10]. Diverse input FGSM (DIM) [16] introduces stochastic perturbations during iterative optimization to mitigate overfitting to the source model. Building upon momentum-based frameworks like momentum iterative FGSM (MI-FGSM) [31] and Nesterov iterative FGSM (NI-FGSM) [17], variance tuning MI-FGSM (VMIM) [18] and variance tuning NI-FGSM (VNIM) [18] dynamically adjust gradient directions using historical variance estimates, thereby escaping local optima while preserving perturbation transferability. Ensemble-driven approaches effectively aggregate knowledge across multiple models or iterations. Large geometric vicinity (LGV) [32] optimizes adversarial examples within high-confidence geometric regions, transferable adversarial attack based on integrated gradients (TAIG) [33] synthesizes perturbations through integrated gradient trajectories, and adaptive model ensemble adversarial attack (AdaEA) [34] adaptively weights for substitute models during attack examples generation.

2.3. Input Transformation Defenses

Input transformation defense mechanisms mitigate adversarial attacks by altering input space characteristics while preserving semantic content. Pixel deflection [35] redistributes pixel distributions coupled with wavelet-domain thresholding. Mixup inference [24] further dilutes adversarial perturbation via linear interpolations with benign examples. JPEG compression [36] eliminates high-frequency adversarial components via quantization, whereas Gaussian blurring [37] suppresses perturbations through spatial smoothing kernels. Advanced neural solutions include feature denoising [38], which embeds denoising blocks within network architectures, and high-level representation guided denoiser (HGD) [39], which aligns adversarial ex-

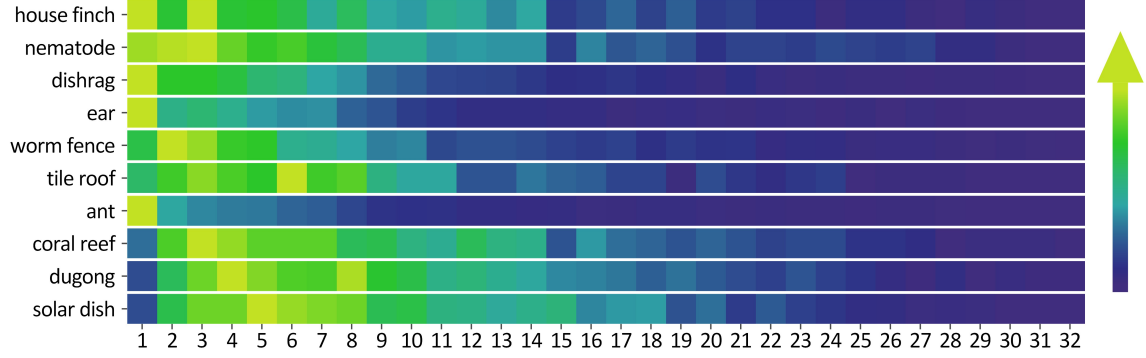


Figure 2. The contribution of different singular value features to adversarial attacks on the CIFAR-10 dataset. A higher intensity in the heatmap indicates a greater contribution.

amples with clean data distributions through loss recalibration. Learning defense transformation (LDT) [40] parameterizes affine transformations using neural boundary information to effectively defend against various attacks.

3. Motivation

For any $m \times n$ matrix A , its singular values are defined as the square roots of the eigenvalues of the matrix $A^T A$. Let the singular values of matrix A be denoted as $\sigma_1, \sigma_2, \dots, \sigma_r$, where r is the rank of matrix A ; then these singular values satisfy the relation $\sigma_1 > \sigma_2 > \dots > \sigma_r > 0$. SVD is a matrix factorization technique in linear algebra, which can decompose matrix A into three matrices:

$$A = U\Sigma V^T, \quad (2)$$

where U is an $m \times m$ unitary matrix (a complex square matrix that satisfies $UU^H = I$). Σ is an $m \times n$ matrix that contains the singular values of A only on its diagonal, while all other elements are zero. V is an $n \times n$ unitary matrix.

The applications of SVD in image processing include image compression, denoising, inpainting, and feature extraction. For instance, in image compression, SVD approximates the original image using a low-rank matrix by discarding smaller singular values, thus significantly reducing storage space. This approach is akin to JPEG compression and has been applied in adversarial attack defenses [29, 30, 36]: specifically, these methods disrupt the structural integrity of adversarial perturbations by removing certain image components, achieving adversarial defense. Due to the limited defensive performance of previous methods, we pose the following questions:

Which image components should be discarded? Should these image components indeed be discarded?

To address these questions, we employed a pre-trained ResNet-18 [4] model as our target model and conducted validation experiments on the CIFAR-10 [41] dataset. We utilized the FGSM [10] method with a perturbation bud-

get of $\ell_\infty = 8/255$ to generate adversarial examples, selecting those that successfully attacked the target model for SVD analysis. We removed the first singular value from the Σ matrix and reconstructed new adversarial examples, then reinserted it and removed the second singular value, repeating this process until the last singular value was assessed. Subsequently, we input all reconstructed adversarial examples into the target model for prediction, recording the number of correct predictions for each category.

We grouped examples by their true categories and generated a heatmap illustrating the contribution of singular value features to the adversarial attacks, as shown in Fig. 2. A higher color intensity in the heatmap indicates that the singular value feature has a greater contribution to the adversarial attack. Overall, features corresponding to larger singular values (on the left) contribute more significantly to the attack, while smaller ones (on the right) have less impact. However, the contributions of identical singular value features vary across categories. For instance, the first singular value feature has the highest contribution in the “dishrag,” “ear,” and “ant” categories, while the second, sixth, and third singular value features dominate in the “worm fence,” “tile roof,” and “coral reef” categories, respectively.

The aforementioned validation experiment indicates the difficulty in determining which singular value features should be removed. Fixed removal may yield positive results for some examples while negatively affecting others. Thus, a flexible approach for adjusting these features is essential. In the next section, we propose a method that utilizes deep learning to reconstruct and optimize singular value features for more effective adversarial defense.

4. Methodology

In this section, we propose a plug-and-play *SVD-driven feature evolution module* (SDFEM) to assist the target model in defending against transfer-based adversarial attacks. The SDFEM is positioned at the input side of the

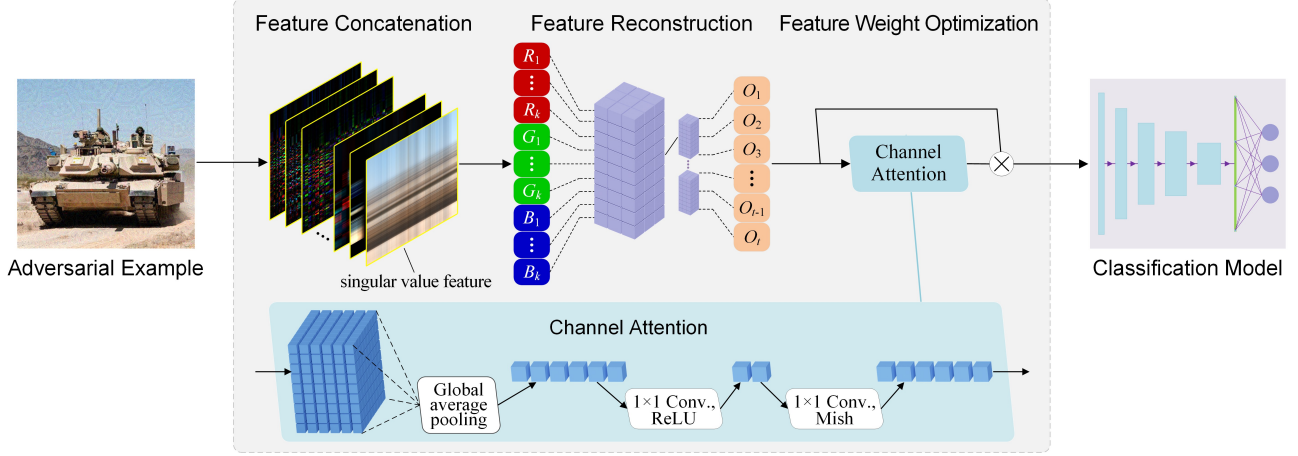


Figure 3. The architecture of SVD-driven feature evolution module (SDFEM).

target model and processes the singular value features obtained from SVD decomposition. Through a series of reconstruction and optimization operations, it suppresses those features that significantly contribute to adversarial attacks.

During the training phase, a mixture of clean examples and adversarial examples generated by a pre-trained model is used as the training data, which is processed by the SDFEM before being fed into the target model. In this phase, the computation graph and gradient information are retained, allowing the parameters of both the SDFEM and the target model to be updated based on the backpropagation algorithm. The input example processing procedure during the inference phase is similar to that of the training phase, with the distinction that the computation graph and gradient information are not retained.

As shown in Fig. 3, the SDFEM consists of three components: “feature concatenation,” “feature reconstruction,” and “feature weight optimization.” We will proceed to introduce each of these components in turn. The algorithm description of SDFEM can be found in Algorithm 1.

4.1. Feature Concatenation

Eq. (2) represents the product form of the SVD, while its summation form can be expressed as shown in Eq. (3).

$$A = \sum_{i=1}^r A_i = \sum_{i=1}^r \sigma_i u_i v_i^T. \quad (3)$$

In Eq. (3), u_i and v_i denote the i -th column vectors of the square matrices U and V , respectively, referred to as the left singular vector and right singular vector. The product $A_i = \sigma_i \cdot u_i \cdot v_i^T$ is defined as the i -th singular value feature.

In the feature concatenation component, we first apply SVD to decompose the input example into k singular value features, where $k = \max(m, n)$. If $r < k$, the missing features are supplemented with zeros. Subsequently, these

singular value features are concatenated along the channel dimension. If the input example is a color image, the image dimensions transform from $(3, m, n)$ to $(3k, m, n)$.

The purpose of feature concatenation is to convert the input examples into a set of singular value features via SVD, facilitating subsequent processing operations.

4.2. Feature Reconstruction

In the feature reconstruction component, we configure a convolutional layer with an input channel of $3k$, an output channel of t , a kernel size of 3×3 , a stride of 1, a padding of 1, and no bias. Following this convolutional layer is a batch normalization (BN) layer with t feature channels. The concatenated singular value features are initially processed by the convolutional layer, then fed into the BN layer for normalization, ultimately resulting in the output of the reconstructed features for subsequent processing components.

The feature reconstruction component is designed to merge and deepen various singular value features, thereby obtaining a multi-level feature representation.

4.3. Feature Weight Optimization

In the feature weight optimization component, we designed a residual connection and a channel attention mechanism. The channel attention mechanism consists of a global average pooling operation, a first convolutional layer, a rectified linear unit (ReLU) activation function, a second convolutional layer, and a Mish activation function. The first and second convolutional layers are configured with an input channel count of t and $t/2$, respectively, and an output channel count of $t/2$ and t , respectively. The remaining parameters include a kernel size of 1×1 , a stride of 1, zero-padding, and the use of a bias.

The reconstructed singular value features are initially compressed into a global average value through a global average pooling operation along the channel dimension.

Algorithm 1 The algorithm description of SDFEM.

Require: The input example for classification

Ensure: The optimized singular value features

- 1: Apply SVD to the input example to obtain singular value features, and subsequently concatenate these features along the channel dimension;
 - 2: Input the concatenated singular value features into a convolutional layer for feature reconstruction;
 - 3: Feed the reconstructed singular value features into the channel attention mechanism to get feature weights;
 - 4: Multiply the feature weights with the reconstructed singular value features to optimize the latter, thereby suppressing those features that contribute significantly to adversarial attacks;
 - 5: Output the reconstructed and optimized singular value features as input for the classification model.
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Subsequently, the first convolutional layer of the attention mechanism and the ReLU function aggregate spatial information to obtain channel-level global context. Following this, the second convolutional layer and the Mish function dynamically optimize the feature weights based on the global context dependencies, enhancing important features while suppressing redundant information. Finally, the reconstructed singular value features are multiplied by the optimized feature weights through a residual connection, resulting in the optimized singular value features that serve as input for the subsequent classification model.

The aim of feature weight optimization is to adjust the reconstructed singular value features by optimizing the weights of different features, thus suppressing those ones that contribute more significantly to adversarial attacks.

5. Experiments

In this section, we first introduce various experimental parameters in Subsec. 5.1. Then, in Subsec. 5.2 and 5.3, we evaluate the defense effectiveness of different methods against unknown types of transfer-based attacks and integrated transfer-based attacks, respectively. Subsequently, we conduct an ablation study on the proposed methods in Subsec. 5.4 to assess the effectiveness of each component. In Subsec. 5.5, we provide a visualization of the defense effectiveness against transfer-based attacks to more intuitively demonstrate the advantages of the proposed methods. Finally, we discuss the limitations of the proposed methods and potential improvements for the future in Subsec. 5.6.

5.1. Experimental Setup

Datasets & Classification models. We use CIFAR-10 [41] and Mini-ImageNet [42] as the training and evaluation datasets. WRN-28-10 [43] and TinyViT-21M [44] are used

as the target models subjected to attacks.

Competitive Methods. We employ natural training, adversarial training [12], feature denoising [38], pixel deflection [35], mixup inference [24], HGD [39] and LDT [40] as competitive methods. These methods utilize the same target models and training parameters as SDFEM.

Training details. With the exception of natural training, which uses clean examples as training data, all other defense methods employ a mixture of adversarial and clean examples as training data. Adversarial examples are generated on a pre-trained ResNet-18 [4] using the PGD with a perturbation budget of $\ell_\infty = 8/255$, a step size of $2/255$, and an iteration count of 20. The optimizer, initial learning rate and weight decay for all defense methods are set to SGD (a momentum of 0.9), 0.01, and 0, respectively. The number of training epochs is set to 120, with the learning rate decreasing by a factor of 10 at the 50th, 75th, and 100th epochs. The batch size of the dataset is set to 256.

Evaluation details. During the evaluation phase, we employ FGSM [10], BIM [14], DIM [16], VMIM [18], VNIM [18], LGV [32], TAIG [33], and AdaEA [34] as attack methods to generate adversarial examples on substitute models based on transferability. These methods are not used in the training process of all defense methods. The perturbation budget for all attack methods is set to $\ell_\infty = 8/255$. For iterative attack methods, the number of iterations is 100 with a step size of $2/255$, while the remaining parameters adhere to the default settings of the *torchattacks* library [45].

5.2. Defending Against Unknown Types of Transfer-Based Attacks

We evaluate the effectiveness of various defense methods against unknown types of transfer-based adversarial attacks. ResNet-34 and ShuffleNet-V2- $2\times$ serve as substitute models for launching the attacks, respectively. The evaluation datasets are CIFAR-10 and Mini-ImageNet. The evaluation results using WideResNet (WRN)-28-10 as the target model are presented in Tab. 1, while the results using TinyViT-21M as the target model are shown in Tab. 2.

In Tab. 1, it can be found that regardless of the defense strategy employed, there will be a reduction in the target model’s classification accuracy on clean examples. In comparison, SDFEM shows the least degradation of clean classification accuracy. When faced with unknown types of transfer-based attacks, SDFEM consistently demonstrates superior defensive performance compared to other defense methods, always exhibiting optimal performance against each type of attack. Its average defense performance on CIFAR-10 and Mini-ImageNet is 95.48% and 71.54%, respectively 1.54% and 2.54% higher than LDT.

In Tab. 2, we employ the Transformer-based TinyViT-21M as the target model. Despite the significant structural differences between this model and ResNet-34 as well as

Table 1. Accuracy comparison (%) of different methods in defending against unknown types of transfer-based attacks on CIFAR-10 and Mini-ImageNet. WRN-28-10 serves as the target model under attack, while ResNet-34 and ShuffleNet-V2-2 $\times$ serve as substitute models for launching the attacks. The best results are boldfaced, and the second best results are underlined.

Dataset	Method	Clean	ResNet-34					ShuffleNet-V2-2 $\times$				
			FGSM	BIM	DIM	VMIM	VNIM	FGSM	BIM	DIM	VMIM	VNIM
CIFAR-10	Nat. Tra.	96.74	28.26	0.65	0.76	0.21	0.18	51.84	10.45	6.88	4.04	3.61
	Adv. Tra.	89.07	87.54	88.83	88.15	87.38	86.98	86.10	86.75	86.33	85.57	85.49
	Feat. Den.	89.62	88.00	88.98	88.45	87.90	87.82	86.08	87.06	86.48	85.57	86.06
	Pix. Def.	92.14	90.29	91.86	91.43	91.38	91.03	88.54	90.23	90.00	89.68	90.06
	Mix. Inf.	95.92	92.43	94.15	<u>93.91</u>	93.23	92.99	89.41	<u>94.43</u>	91.86	92.91	92.54
	HGD	94.24	93.13	94.01	93.66	93.98	93.89	92.64	92.73	92.62	<u>93.80</u>	<u>93.82</u>
	LDT	96.03	<u>94.59</u>	<u>94.80</u>	93.88	93.13	93.39	<u>93.87</u>	93.86	<u>93.76</u>	93.18	92.83
	★ SDFEM	<u>96.17</u>	95.23	95.71	94.75	95.15	94.98	96.01	95.45	95.48	95.51	95.80
Mini-ImageNet	Nat. Tra.	77.19	12.40	0.47	0.36	0.28	0.15	18.51	2.62	1.93	0.40	0.35
	Adv. Tra.	63.28	61.97	62.51	61.80	61.80	61.67	57.01	57.05	56.53	56.38	56.25
	Feat. Den.	63.81	59.24	59.50	56.92	56.61	56.45	53.79	54.10	51.71	50.86	51.33
	Pix. Def.	68.28	66.08	67.91	66.24	66.17	65.94	60.34	62.77	60.91	60.88	60.76
	Mix. Inf.	74.49	69.35	68.81	<u>68.97</u>	68.31	<u>69.92</u>	64.14	65.65	65.48	64.75	65.89
	HGD	73.05	70.10	69.79	67.78	67.91	68.42	<u>67.56</u>	64.94	65.98	65.94	64.82
	LDT	74.40	<u>71.54</u>	<u>70.14</u>	68.29	<u>69.81</u>	69.73	66.96	<u>67.07</u>	<u>66.95</u>	<u>66.81</u>	<u>67.39</u>
	★ SDFEM	<u>75.48</u>	73.63	72.24	70.84	71.70	71.43	69.81	71.18	69.74	70.27	70.58

Table 2. Accuracy comparison (%) of different methods in defending against unknown types of transfer-based attacks on CIFAR-10 and Mini-ImageNet. TinyViT-21M serves as the target model under attack, while ResNet-34 and ShuffleNet-V2-2 $\times$ serve as substitute models for launching the attacks. The best results are boldfaced, and the second best results are underlined.

Dataset	Method	Clean	ResNet-34					ShuffleNet-V2-2 $\times$				
			FGSM	BIM	DIM	VMIM	VNIM	FGSM	BIM	DIM	VMIM	VNIM
CIFAR-10	Nat. Tra.	96.36	28.46	0.42	0.36	0.45	0.25	51.81	10.00	7.09	3.71	3.79
	Adv. Tra.	88.77	87.54	87.84	87.79	86.94	86.88	86.25	86.42	86.83	85.43	85.46
	Feat. Den.	89.26	88.04	89.05	88.11	88.26	87.45	85.60	87.22	86.27	85.12	86.48
	Pix. Def.	93.83	90.23	92.00	91.55	91.82	91.56	88.89	90.06	89.91	89.70	89.66
	Mix. Inf.	95.46	92.05	93.07	92.98	<u>93.72</u>	93.43	89.47	<u>93.83</u>	91.82	92.74	92.79
	HGD	94.38	93.12	92.41	93.21	93.34	<u>93.69</u>	93.00	92.34	92.89	<u>93.47</u>	<u>94.06</u>
	LDT	95.81	<u>93.95</u>	<u>93.25</u>	<u>93.32</u>	93.42	93.18	<u>93.72</u>	93.35	<u>93.02</u>	92.89	92.89
	★ SDFEM	<u>96.01</u>	95.17	95.09	94.98	95.16	94.81	94.96	95.49	95.08	94.92	94.48
Mini-ImageNet	Nat. Tra.	79.59	14.56	3.47	2.81	2.29	2.75	21.18	5.46	4.46	3.03	2.40
	Adv. Tra.	65.23	63.49	64.51	63.43	63.74	64.23	59.01	59.75	58.22	58.76	58.30
	Feat. Den.	65.51	62.09	61.44	58.54	59.23	59.12	56.20	56.70	54.30	53.47	53.86
	Pix. Def.	70.10	67.87	69.50	67.91	67.68	67.87	62.95	64.34	62.57	62.73	63.16
	Mix. Inf.	<u>77.34</u>	70.99	70.76	69.83	69.88	69.46	65.79	67.26	66.15	67.35	67.58
	HGD	74.65	71.06	70.35	69.08	68.41	68.76	<u>68.25</u>	66.57	65.41	<u>67.90</u>	67.44
	LDT	75.26	<u>72.72</u>	<u>71.40</u>	<u>69.95</u>	<u>70.11</u>	<u>70.32</u>	67.28	<u>69.77</u>	<u>68.97</u>	67.61	<u>68.03</u>
	★ SDFEM	77.06	75.28	74.88	73.58	73.10	73.22	72.04	73.10	72.22	71.90	72.21

ShuffleNet-V2-2 $\times$, SDFEM still demonstrates the highest defensive performance against transfer-based attacks initiated using these substitute models. Its average defense performance on CIFAR-10 and Mini-ImageNet is 95.10% and 73.51%, respectively 1.57% and 3.38% higher than LDT.

5.3. Defending Against Integrated Transfer-Based Attacks

We evaluated the effectiveness of different defense methods against integrated adversarial attacks. The integrated attack methods include LGV [32], TAIG [33], and AdaEA [34]. The basic attack method is BIM. For LGV, the sub-

stitute model is ResNet-34 and the number of weight sets is 20. For TAIG, the substitute model is also ResNet-34, and the example augmentation factor is 20. For AdaEA, the substitute model ensemble consists of ResNet-34, ShuffleNet-V2-2 $\times$, and VGG-19. For all defense methods, these substitute models have never been encountered. The evaluation results using WRN-28-10 and TinyViT-21M as target models on the CIFAR-10 and Mini-ImageNet datasets are presented in Tab. 3. It is evident that when confronted with integrated attacks involving multiple gradients or models, SDFEM continues to exhibit the highest defensive performance compared to all other defense methods.

Table 3. Accuracy comparison (%) of different methods in defending against integrated transfer-based attacks on CIFAR-10 and Mini-ImageNet. WRN-28-10 serves as the target model. The best results are boldfaced, and the second best results are underlined.

Method	CIFAR-10			Mini-ImageNet		
	LGV	TAIG	AdaEA	LGV	TAIG	AdaEA
Nat. Tra.	22.77	1.78	0.11	8.43	0.77	0.42
Adv. Tra.	85.20	84.10	84.79	55.66	54.39	54.52
Feat. Den.	86.39	85.29	84.29	56.11	54.27	54.73
Pix. Def.	88.55	86.23	86.60	60.94	58.03	57.53
Mix. Inf.	87.23	86.53	87.97	63.18	62.68	63.10
HGD	90.64	89.09	90.70	65.26	64.43	63.63
LDT	<u>91.48</u>	<u>90.40</u>	90.43	<u>66.80</u>	<u>65.45</u>	<u>64.12</u>
★ SDFEM	94.15	93.10	93.76	69.49	69.61	68.13

Table 4. Accuracy comparison (%) against different transfer-based adversarial attacks in the ablation study of SDFEM. WRN-28-10 serves as the target model. The best results are boldfaced, and the second best results are underlined.

NO.	Con.	Rec.	WOpt.	Clean	BIM	DIM	VMIM	VNIM
①	×	×	×	89.07	88.83	88.15	87.38	86.98
②	✓	×	×	92.25	90.77	90.53	91.44	90.40
③	×	✓	×	91.32	88.75	89.04	87.51	87.13
④	×	×	✓	90.88	88.95	88.58	88.19	86.63
⑤	✓	✓	×	94.08	92.72	93.31	92.29	91.71
⑥	✓	×	✓	<u>94.12</u>	<u>94.31</u>	<u>93.49</u>	<u>93.03</u>	<u>92.68</u>
⑦	×	✓	✓	92.03	90.49	89.56	89.39	87.54
⑧	✓	✓	✓	96.17	95.71	94.75	95.15	94.98

5.4. Ablation Study

The SDFEM is divided into three components: “feature **C**oncatenation,” “feature **R**econstruction,” and “feature **W**eight **O**ptimization.” We conducted an ablation study on CIFAR-10 using WRN-28-10 and ShuffleNet-V2-2× as the target model and the substitute model, respectively. The results are presented in Tab. 4. To accurately assess the contribution of each component to the defensive performance, we computed their average marginal effects across all possible combinations, akin to the Shapley value [46] in cooperative game theory. The contributions of the above three components to the accuracy of defending against transfer-based attacks were found to be +4.37%, +1.35%, and +1.65%, respectively. This indicates that “feature concatenation” is the most critical component of the SDFEM, followed by “feature weight optimization” and “feature reconstruction.”

5.5. t-SNE Visual Analysis

We input clean and adversarial examples into both the undefended WRN-28-10 and the SDFEM-defended WRN-28-10, followed by visualization analysis of the output predictions using t-SNE (t-distributed Stochastic Neighbor Embedding) [47], as shown in Fig. 4. The adversarial examples were generated using the PGD with ShuffleNet-V2-

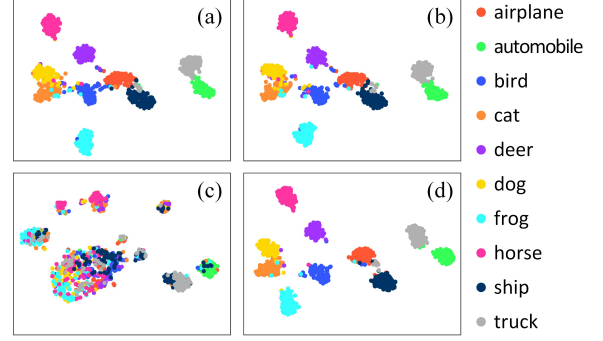


Figure 4. t-SNE visualization results on CIFAR-10. Visualization of clean examples on (a) the undefended model and (b) the SDFEM-defended model; visualization of adversarial examples on (c) the undefended model and (d) the SDFEM-defended model.

2× as a substitute model. It is observed that the clustering results for clean examples remain clear before and after defense, indicating that SDFEM rarely misclassifies them. The clustering results for adversarial examples from the model without defenses are highly chaotic, suggesting its inability to classify them correctly. In contrast, SDFEM achieves remarkably clear clustering results for them, demonstrating its strong adversarial defense capability.

5.6. Limitations and Future Work

Due to the linear superposition relationship between the original images inputted in the “feature concatenation” part and the singular value features outputted, this segment is capable of propagating gradients. Additionally, both “feature reconstruction” and “feature weight optimization” are the convolution operations, which also facilitate gradient flow. Consequently, SDFEM is primarily suited for defending against transfer-based adversarial attacks in black-box scenarios, while it struggles to defend against gradient-based adversarial attacks in white-box scenarios. In future work, we intend to explore adaptive adversarial training methods for SDFEM based on several advanced white-box defense techniques [48, 49], aiming to enhance its robustness against gradient-based adversarial attacks.

6. Conclusion

In this paper, we reveal that the contributions of singular value features to adversarial attacks vary across different categories, making it challenging to quantitatively assess them. To address this issue, we propose a plug-and-play module named SDFEM to suppress the singular value features that significantly contribute to adversarial attacks, thereby assisting target classification models in defending against transfer-based adversarial attacks. Experiments demonstrate that the proposed method exhibits outstanding robustness in black-box attack scenarios.

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